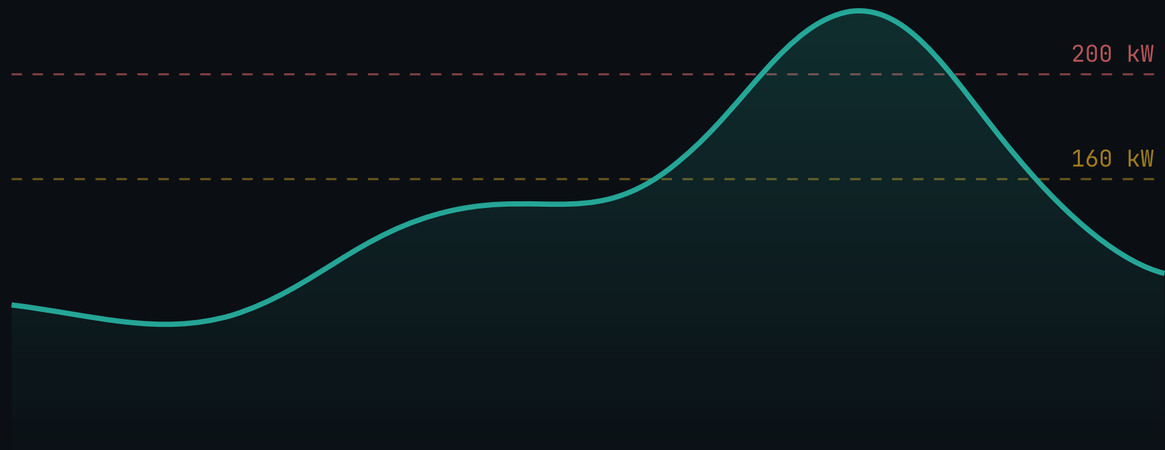
NO SUS.

GridPilot

Forecast tomorrow's load.
Switch before it spikes.



HackSphere 3.0 · BITS Pilani Hyderabad · Team NO SUS

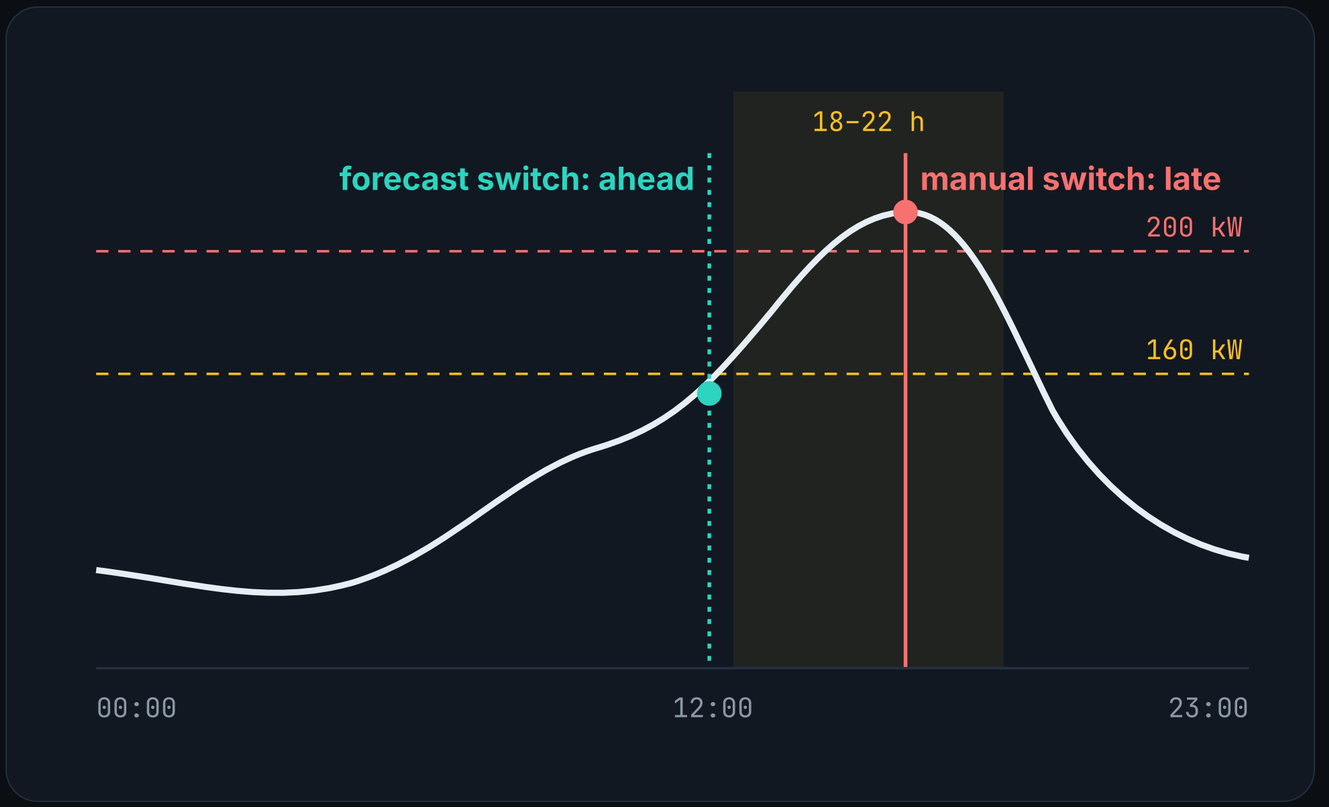
`live app` https://gridpilot-50047251149.development.catalystappsail.in

Every hour, a microgrid must pick: main grid or battery.

24 / day
source decisions, one per hour

18–22 h
evening peak, every single day

≥ 200 kW
heat waves push load into overload



One pipeline. Weather in, signals out.

01
Ingest
Live form or CSV upload

→

02
Clean
Validate, fill gaps, build features

→

03
Forecast 24 h
QuickML endpoint, one kW value per hour

→

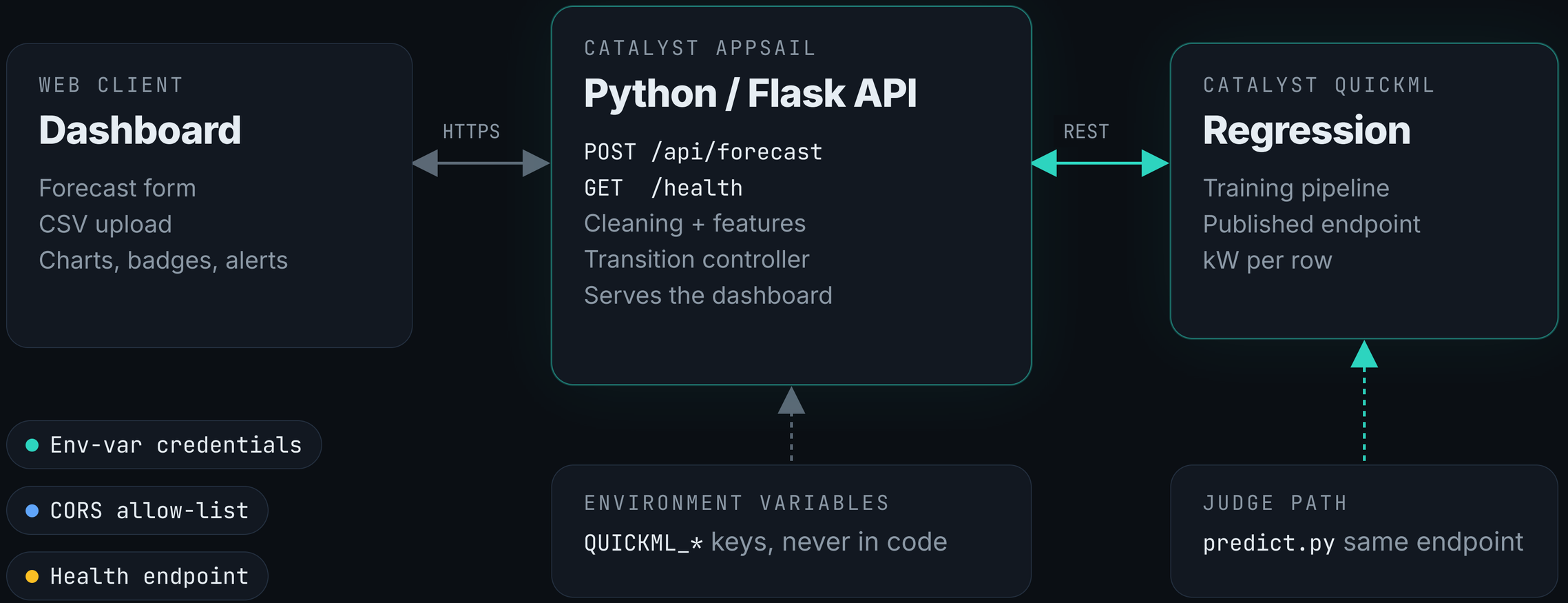
04
Controller
Grid or storage, plus overload alert

→

05
Dashboard + signals
Load curve, mode timeline, alerts, log

Every forecast hour gets a **mode** and an **alert state** before that hour arrives.

Three parts. All on Catalyst.



4,000 training hours. Cleaned the same way at train and predict time.

SCHEMA

timestamp	YYYY-MM-DD HH
ambient_temp_c	°C
humidity_pct	%
solar_irradiance_wm2	W/m²
→ target_demand_kw	kW

CLEANING

- **Normalize timestamps**
  parse, floor to hour, sort
- **Interpolate gaps**
  time-based, edges filled
- **Clip outliers**
  physical ranges per sensor
- **Dedupe**
  one row per hour

FEATURES

- **Hour, weekday**
- **Cyclic hour**
  sin / cos
- **Cooling / heating degrees**
- **Evening window flag**
  18–22 h

Ridge + LightGBM residual hybrid

VALIDATION RMSE

9.70 kW

VALIDATION MAE

7.74 kW

CHRONOLOGICAL HOLD-OUT

train 80% | 20%

Split by time, not shuffled. No future rows leak into training.

The test window is summer.

Hotter than most training hours. Tree models cannot extrapolate past what they saw.

So: linear + boosted residuals.

Linear temperature terms carry the heat trend. Gradient boosting learns what is left.

Three rules. Every edge tested.

Demand > 160 kW
or hour in 18:00–22:00 → **Renewable Storage**

Otherwise → **Main Grid**

Demand ≥ 200 kW
independent of mode → **Critical Overload Alert**

BOUNDARY TESTS

160.00 kW	Main Grid	✓
160.01 kW	Storage	✓
200.00 kW	Alert	✓
22:59	Storage	✓
23:00	Main Grid	✓

See the next 24 hours at a glance.



Desktop · live Catalyst app



Mobile · responsive

- **Load curve**
  160 / 200 kW lines
- **24 h mode timeline**
  grid or storage
- **Battery SoC**
- **Alert banners**
  ≥ 200 kW
- **Dispatch signal log**

Built for the hidden test. Runs first time.

01 One row in, one prediction out
Timestamp kept, order kept

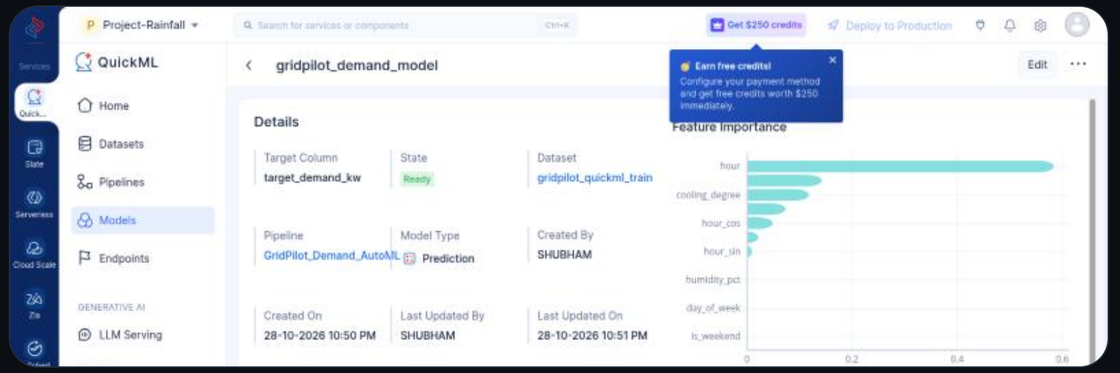
02 Chunked calls
Large files sent in batches

03 Retry
Backoff on timeouts and 5xx

04 Fallback
Never an empty or short CSV

- Public QuickML endpoint
- README: setup, architecture, env vars

```
$ python predict.py \
    --input sample_test_input.csv \
    --output predictions.csv
$ python test_eval.py
SUCCESS
```



QuickML pipeline · published endpoint

From forecast to live control.

01

Real meter feeds

Telemetry pushed in, not uploaded

`Catalyst Event Listeners`

02

Forecast history

Predicted vs actual, tracked over time

`Catalyst Data Store`

03

Per-site models

One model per microgrid, its own weather

`QuickML pipelines`

04

SMS alerts

Operators warned before the overload hour

`Catalyst Signals`

NO SUS.

Shubham Sahu · Team leader

B.E. Electronics & Computer Engineering · TSEC Mumbai

LIVE APP	https://gridpilot-500472511149.development.catalystappsail.in
CODE	https://github.com/https-shubhamsahu/hacksphere-gridpilot
GITHUB	github.com/https-shubhamsahu

Thank you.